

Motherboards

The Lifeline of Your Desktop

If you are a gamer, or like to customise your PC and build it from scratch, the motherboard is what you require to link all the important and key components together. Let's find out how to select the best desktop motherboards.

The central processing unit (CPU) can be considered to be the brain of a system or a PC in layman's language, but it still needs a 'nervous system' to be connected with all the other components in your PC. A motherboard plays this role, as all the components are attached to it and to each other with the help of this board. It can be defined as a PCB (printed circuit board) that has the capability of expanding. As the name suggests, a motherboard is believed to be the 'mother' of all the components attached in it, including network cards, sound cards, hard drives, TV tuner cards, slots, etc. It holds the most significant sub-systems—the processor along with other important components. A motherboard is found in all electronics devices like TVs, washing machines and other embedded systems. Since it provides the electrical connections through which other components are connected and linked with each other, it needs the most attention. It hosts other devices and subsystems and also contains the central processing unit, unlike the backplane.

There are quite a lot of companies that deal with motherboards and Simmtronics is one among the leading players. According to Dr Inderjeet Sabbrawal, chairman, Simmtronics, "Simmtronics has been one of the exclusive manufacturers of motherboards in the hardware industry over the last 20 years. We strongly believe in creativity, innovation and R&D. Currently, we are fulfilling our commitment to provide the latest mainstream motherboards. At Simmtronics, the quality of the motherboards is strictly controlled. At present, the market is not growing.... India still has a varied market for older generation models as well as the latest models of motherboards."

Factors to consider while buying a motherboard

In a desktop, several essential units and components are attached directly to the motherboard, such as the microprocessor, main memory, etc. Other components, such as the external storage controllers for sound and video display and various peripheral devices, are attached to it through slots, plug-in cards or cables. There are a number of factors to keep in mind while buying a motherboard, and these depend on the specific requirements. Linux is slowly taking over the PC world and, hence, people now look for Linux-supported motherboards. As a result, almost every motherboard now supports Linux. The many factors to keep in mind when buying a Linux-supported motherboard are discussed below.

CPU socket

The central processing unit is the key component of a motherboard and its performance is primarily determined by the kind of processor it is designed to hold. The CPU socket can be defined as an electrical component that connects or attaches to the motherboard and is designed to house a microprocessor. So, when you're buying a motherboard, you should look for a CPU socket that is compatible with the CPU you have planned to use. Most of the time, motherboards use one of the following five sockets -- LGA1155, LGA2011, AM3, AM3+ and FM1. Some of the sockets are backward compatible and some of the chips are interchangeable. Once you opt for a motherboard, you will be limited to using the processors that offer similar specifications.

Form factor

A motherboard's capabilities are broadly determined by its shape, size and how much it can be expanded — these aspects are known as form factors. Although there is no fixed design or form for motherboards, and they are available in many variations, two form factors have always been the favourites -- ATX and microATX. The ATX motherboard measures around 305cm x 23cm (12 inch x 9 inch) and offers the highest number of expansion slots, RAM bays and data connectors. MicroATX motherboards measure 24.38cm x 24.38cm (9.6 x 9.6 inch) and have fewer expansion slots, RAM bays and other components. The form factor of a motherboard can be decided according to what purpose the motherboard is expected to serve.

RAM bays

Random access memory (RAM) is considered the most important workspace in a motherboard, where data is processed even after being removed from the hard disk drive or solid state drive. The efficiency of your PC directly depends on the speed and size of your RAM. The more space you have on your RAM, the more efficient your computing will be. But it's no use having a RAM with greater efficiency than your motherboard can support, as that will be just a waste of the extra potential. Neither can you have RAM with lesser efficiency than the motherboard, as then the PC will not work well due to the bottlenecks caused by mismatched capabilities. Choosing the motherboard which supports just the right RAM is vital.

Apart from these factors, there are many others to consider before selecting a motherboard. These include the audio system, display, LAN support, expansion capabilities and peripheral interfaces.